Unit III: Calculus and Optimization

Derivatives, partial derivatives, gradient, optimization, maxima and minima

Learning Goals

- Understand derivatives as rates of change and slopes of curves.
- Use partial derivatives to study functions with many variables.
- Interpret the gradient as the direction of steepest increase.
- Apply optimization ideas to minimize loss functions and maximize useful objectives.
- Identify local maxima, local minima, and saddle points using first and second derivative ideas.
- Connect calculus and optimization to machine learning, data fitting, and quantum systems.

1. Why Calculus Matters

Calculus studies change. In artificial intelligence, learning usually means changing model parameters so that predictions become better. In quantum systems, calculus appears when studying continuous changes in states, dynamics, control pulses, and optimization of experiments.

Optimization is the practical side of calculus. Instead of only asking how a function changes, optimization asks how to choose inputs that make a function as small or as large as possible. Training a model, fitting a curve, reducing error, and tuning a quantum circuit are all optimization problems.

2. Functions and Rates of Change

A function maps inputs to outputs. For example, f(x) = x^2 maps x to its square. In applications, a function may represent cost, distance, probability, energy, prediction error, or accuracy.

```
Function notation: y = f(x)        Change in output: Delta y = f(x + Delta x) - f(x)
```

The average rate of change over an interval compares how much the output changes relative to the input change. The derivative is the instantaneous rate of change at a point.

```
Average rate of change = [f(x + h) - f(x)] / h
```

3. Derivatives

The derivative of f at x measures the slope of the tangent line to the curve y = f(x). If the derivative is positive, the function is increasing at that point. If it is negative, the function is decreasing. If it is zero, the function may have a maximum, minimum, or flat point.

```
f’(x) = limit as h -> 0 of [f(x + h) - f(x)] / h
```

Derivative sign	Meaning
$f'(x) > 0$	Function is increasing locally.
$f'(x) < 0$	Function is decreasing locally.
$f'(x) = 0$	Possible maximum, minimum, or saddle/flat point.

Common Derivative Rules

```
d/dx [c] = 0        d/dx [x^n] = n x^(n-1)        d/dx [c f(x)] = c f’(x)
```

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d/dx [f(x) + g(x)] = f’(x) + g’(x)        d/dx [e^x] = e^x        d/dx [ln x] = 1/x
```

- Power rule: reduces the power by one and multiplies by the original power.
- Sum rule: derivative of a sum is the sum of derivatives.
- Chain rule: used when one function is inside another function.

```
Chain rule: if y = f(g(x)), then dy/dx = f’(g(x)) g’(x)
```

The chain rule is essential in neural networks. Backpropagation is mainly an organized use of the chain rule to compute how each parameter affects the final loss.

4. Second Derivatives

The second derivative measures how the first derivative changes. It gives information about curvature. A positive second derivative means the graph bends upward. A negative second derivative means the graph bends downward.

```
Second derivative: f’’(x) = d/dx [f’(x)]
```

- If $f''(x) > 0$, the function is locally convex or cup-shaped near x.
- If $f''(x) < 0$, the function is locally concave or cap-shaped near x.
- If $f''(x) = 0$, the test may be inconclusive.

5. Partial Derivatives

Many real functions depend on more than one variable. For example, a model’s loss may depend on thousands or millions of weights. A partial derivative measures the rate of change with respect to one variable while treating the other variables as constants.

```
For f(x, y), partial derivative with respect to x: df/dx, treating y as constant
```

```
For f(x, y), partial derivative with respect to y: df/dy, treating x as constant
```

Example: if f(x, y) = x^2 + 3xy + y^2, then the partial derivative with respect to x is 2x + 3y, and the partial derivative with respect to y is 3x + 2y.

- Partial derivatives help identify which variable has the strongest local effect.
- In machine learning, each model parameter has its own partial derivative of the loss.
- In optimization, all these partial derivatives are collected into the gradient.

6. Gradient

The gradient is a vector made of all first-order partial derivatives. For a function of many variables, the gradient points in the direction of steepest increase. The negative gradient points in the direction of steepest decrease.

```
grad f = [df/dx1, df/dx2, ..., df/dxn]
```

```
For f(x, y), grad f = [df/dx, df/dy]
```

If the gradient is zero, the function has a stationary point. A stationary point may be a local minimum, local maximum, or saddle point.

- Large gradient magnitude means the function changes rapidly.
- Small gradient magnitude means the function changes slowly or is near a stationary point.
- Gradient direction is used to update parameters in optimization algorithms.

7. Optimization

Optimization means finding the best value of a function according to a goal. In minimization, we try to make a function as small as possible. In maximization, we try to make it as large as possible.

```
Minimization: find x that minimizes f(x)       Maximization: find x that maximizes f(x)
```

Term	Meaning
Objective function	The function being minimized or maximized.
Loss function	An error function minimized during model training.
Parameter	A value adjusted by the optimization method.
Constraint	A condition that limits allowed solutions.
Optimum	A best point according to the objective.

In AI, the objective is often a loss function. The model starts with imperfect parameters, computes prediction error, finds gradients, and updates parameters to reduce the loss.

8. Gradient Descent

Gradient descent is a basic optimization method for minimizing a function. It starts at an initial point, computes the gradient, and moves in the opposite direction of the gradient because that direction gives steepest decrease.

```
Update rule: x_new = x_old - eta grad f(x_old)
```

- eta is the learning rate, a positive scalar controlling step size.
- If eta is too small, learning is slow.
- If eta is too large, the method may overshoot or diverge.
- The method repeats until the loss stops improving or the gradient becomes small.

In neural networks, gradient descent and its variants update weights using gradients computed by backpropagation. Variants such as stochastic gradient descent, momentum, RMSProp, and Adam are widely used in practical training.

9. Maxima and Minima

A maximum is a point where the function value is higher than nearby values. A minimum is a point where the function value is lower than nearby values. Local extrema compare nearby points, while global extrema compare all points in the domain.

Point type	Description
Local maximum	Higher than nearby points.
Local minimum	Lower than nearby points.
Global maximum	Highest point in the entire domain.
Global minimum	Lowest point in the entire domain.
Saddle point	Stationary point that is not a local maximum or local minimum.

First Derivative Test

- Find critical points where $f'(x) = 0$ or $f'(x)$ is undefined.
- If $f'(x)$ changes from positive to negative, the point is a local maximum.
- If $f'(x)$ changes from negative to positive, the point is a local minimum.
- If the sign does not change, the point may not be a maximum or minimum.

Second Derivative Test

```
At critical point c: if f''(c) > 0, local minimum; if f''(c) < 0, local maximum
```

- If f''(c) = 0, use another method because the test is inconclusive.
- For functions of many variables, second-order behavior is studied using the Hessian matrix.

10. Convexity and Optimization Intuition

A convex function is shaped so that any local minimum is also a global minimum. This makes optimization easier. Many simple models in machine learning, such as linear regression with squared error, lead to convex optimization problems.

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Convex intuition: a bowl-shaped loss surface has one best bottom point
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- Convex problems are easier because there are no deceptive local minima.
- Non-convex problems, such as deep neural network training, can have many valleys, flat regions, and saddle points.
- Even in non-convex settings, gradient methods often work well in practice with suitable initialization and learning rates.

11. AI Connections

- Derivatives show how changing one input or parameter changes the output.
- Loss functions measure prediction error; optimization tries to reduce this loss.
- Gradients tell a model how to update weights during training.
- Backpropagation uses the chain rule to compute gradients efficiently.
- Gradient descent is used in linear regression, logistic regression, neural networks, and deep learning.
- Maxima and minima help interpret best-fit parameters, decision boundaries, and model performance.

12. Quantum Systems Connections

Calculus and optimization appear in quantum technology when parameters of quantum circuits, pulses, or experiments must be tuned. In variational quantum algorithms, a quantum circuit has adjustable parameters, and a classical optimizer updates those parameters to minimize an energy or cost function.

- Parameterized quantum circuits depend on variables such as rotation angles.
- A cost function may represent energy, error, or distance from a target state.
- Optimization searches for parameter values that produce the desired quantum behavior.
- Gradient-based and gradient-free methods can both be used depending on hardware noise and measurement limits.

```
Variational idea: choose theta to minimize C(theta)
```

Worked Examples

Example 1: Derivative

Let $f(x) = x^2 + 3x + 2$. Then $f'(x) = 2x + 3$. At $x = 1$, $f'(1) = 5$, so the function is increasing at that point.

Example 2: Minimum

Let $f(x) = x^2 - 4x + 7$. First derivative: $f'(x) = 2x - 4$. Set $f'(x) = 0$, so $x = 2$. Second derivative: $f''(x) = 2$, which is positive. Therefore $x = 2$ is a local minimum.

Example 3: Gradient

Let $f(x, y) = x^2 + y^2$. Then grad f = [2x, 2y]. At (3, 4), grad f = [6, 8]. The negative gradient [-6, -8] points toward decreasing f.

Quick Revision Sheet

- A derivative is the instantaneous rate of change of a function.
- A positive derivative means increasing; a negative derivative means decreasing.
- A second derivative describes curvature.
- A partial derivative changes one variable while holding others constant.
- The gradient is the vector of all first partial derivatives.
- The gradient points toward steepest increase; the negative gradient points toward steepest decrease.
- Optimization finds values that minimize or maximize an objective function.
- Gradient descent updates parameters by moving opposite the gradient.
- Critical points occur where the derivative is zero or undefined.
- Maxima, minima, and saddle points are different types of important points.
- AI training is usually an optimization problem over a loss function.
- Quantum variational algorithms optimize circuit parameters to reduce a cost function.

Practice Questions

- Find the derivative of $f(x) = 5x^3 - 2x + 7$.
- For $f(x) = x^2 - 6x + 10$, find the critical point and classify it.
- Explain the meaning of $f'(x) > 0$ in words.
- Find the partial derivatives of $f(x, y) = x^2 + 4xy + y^2$.
- Compute the gradient of $f(x, y) = 3x^2 + 2y^2$ at (1, 2).
- Why does gradient descent move in the negative gradient direction?
- What happens if the learning rate is too large?
- Differentiate between local minimum and global minimum.
- What is a saddle point?
- Explain how derivatives are used in training a neural network.
- Give one example of optimization in quantum technology.
- Why is convexity useful in optimization?